



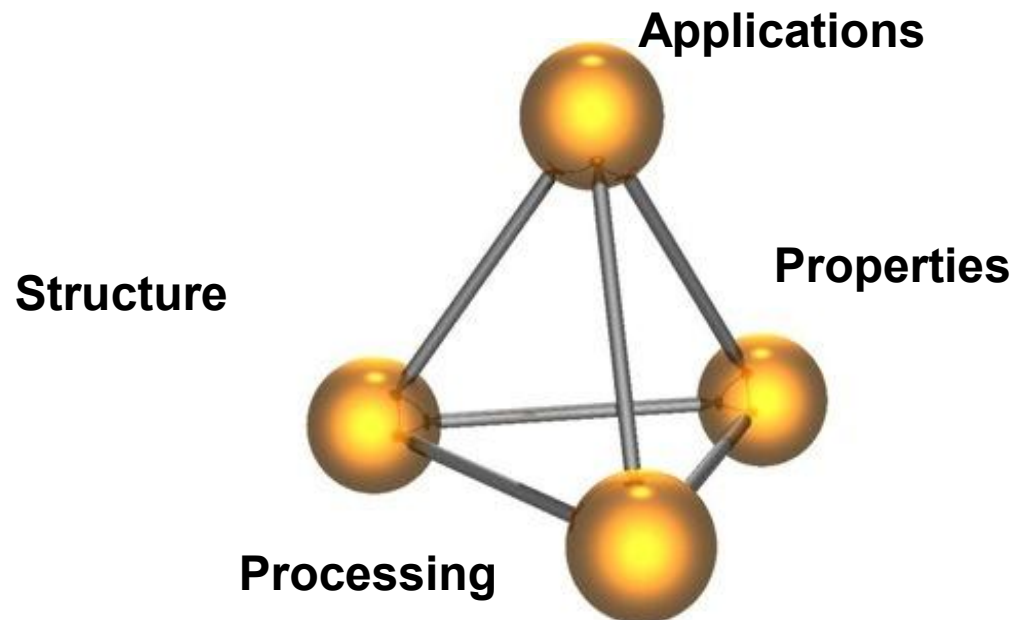
TOPIC 1. BONDING IN SOLIDS

- **Introduction**
- **Bonding in solids**
 - Ionic bonding
 - Covalent bonding
 - Metallic bonding
 - Secondary bonding / Interamolecular forces
- **Bonding and properties in materials**



INTRODUCTION

Nature of bonding $\leftrightarrow$ properties $\leftrightarrow$ type of materials $\leftrightarrow$ Processing and applications



TOPIC 1. BONDING IN SOLIDS

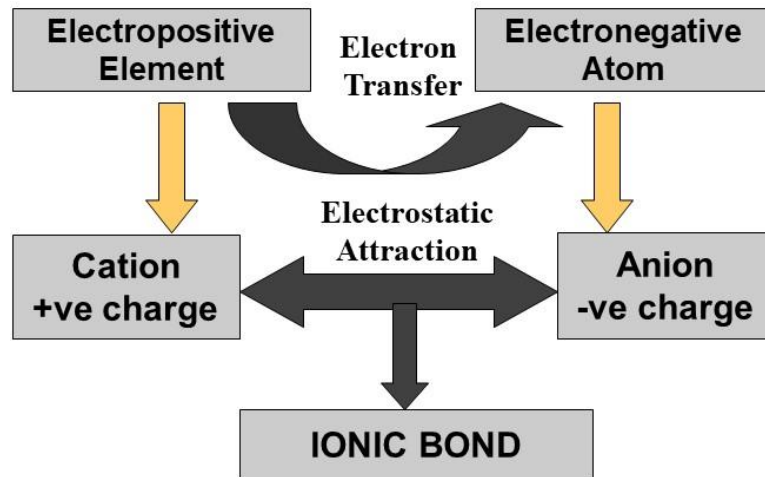
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Ionic Bonding

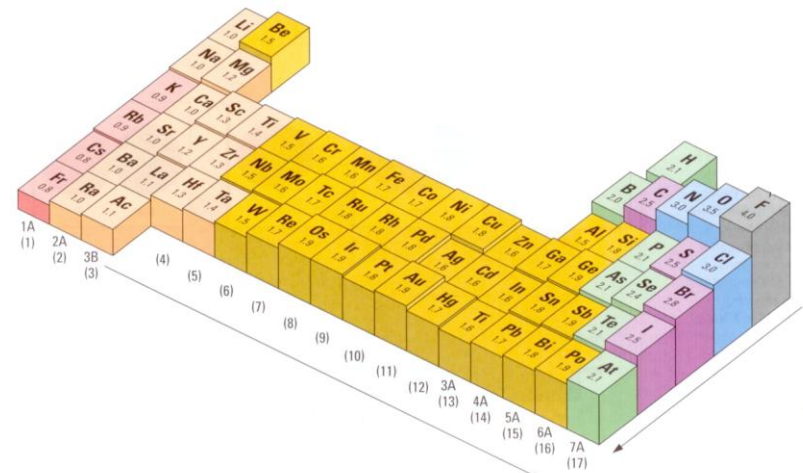
Ionic bonding is the complete transfer of valence electron(s) between atoms. It is a type of chemical bond that generates two oppositely charged ions.

Properties

- ✓ It forms between metal and nonmetallic elements.
- ✓ Electron transfer between the less electronegative atom to the more electronegative
- ✓ Bonding forces: electrostatic attraction between opposite charged ions.
- ✓ Non-directional



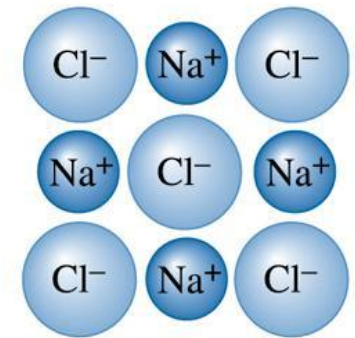
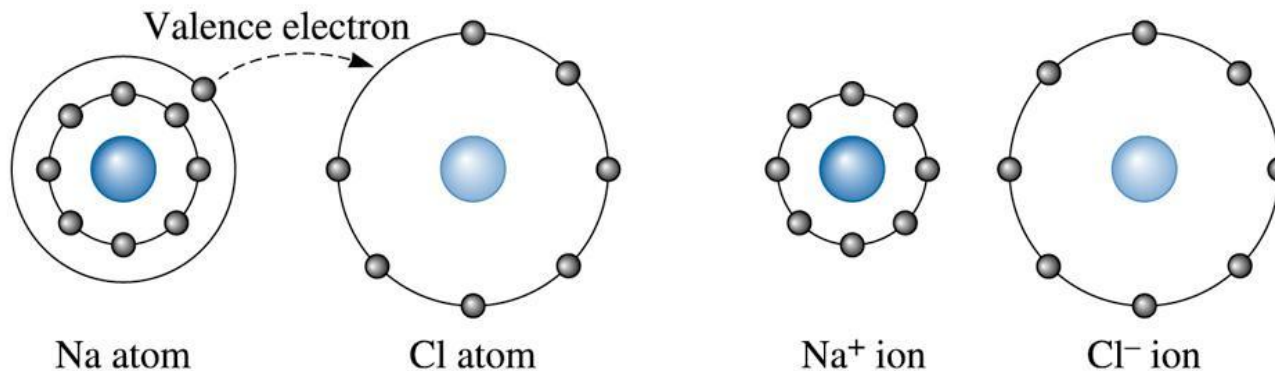
The electronegativity variations



Ionic Bonding

An ionic bond is created between two unlike atoms with different electronegativities : When sodium donates its valence electron to chlorine, each becomes an ion; attraction occurs, and the ionic bond is formed.

- When a metal forms a **cation**, its radius **is reduced** and when a nonmetal forms an **anion**, its **radius increases**



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- Pure ionic bond: ideal.
- ⇒ Always exists covalent participation

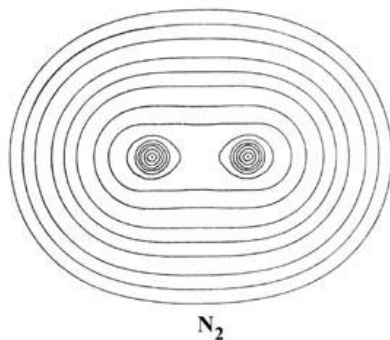
Ionic Bonding

Fundamental principles of ionic structures

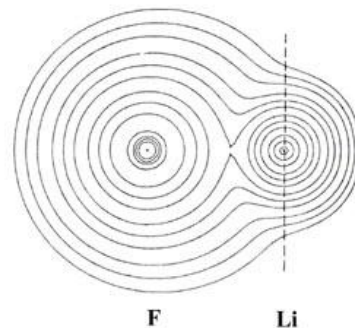
- ✓ Charged spheres
- ✓ Elastic spheres
- ✓ Polarizable (Electron cloud distortion due to the presence of other ions)

Electron density map

Covalent bond



Ionic bond



Ion sizes

Sizes of atoms and their ions in pm

Group 1	Group 2	Group 13	Group 16	Group 17					
Li^+  90	Li  134	Be^{2+}  59	Be  90	B^{3+}  41	B  82	O  73	O^{2-}  126	F  71	F^-  119
Na^+  116	Na  154	Mg^{2+}  86	Mg  130	Al^{3+}  68	Al  118	S  102	S^{2-}  170	Cl  99	Cl^-  167
K^+  152	K  196	Ca^{2+}  114	Ca  174	Ga^{3+}  76	Ga  126	Se  116	Se^{2-}  184	Br  114	Br^-  182
Rb^+  166	Rb  211	Sr^{2+}  132	Sr  192	In^{3+}  94	In  144	Te  135	Te^{2-}  207	I  133	I^-  206

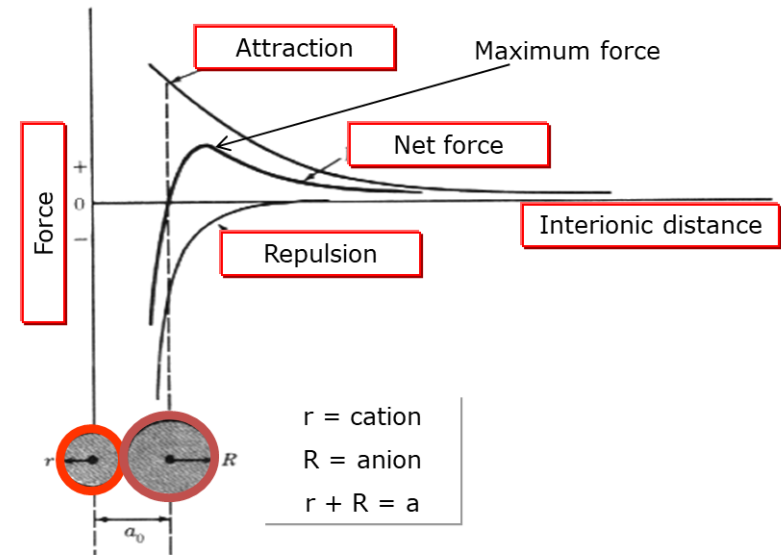
Ionic Bonding

Interionic force and energy for an ionic pair

- ✓ Too far: Distance neither attract nor rep
- ✓ Higher distance: Coulombic attraction
- ✓ Lower distance: Cloud repulsion

$$F_{net} = F_{attractive} + F_{repulsive}$$

$$F_{net} = -\frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 a^2} - \frac{nb}{a^{n+1}}$$



Z_1, Z_2 : number of electrons removed or added
 e : electron charge = $1.6 \times 10^{-19} \text{ C}$
 ϵ_0 : permittivity of space = $8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
 b : constant
 n : 5-9 (Born exponent)
 a_0 : interionic distance

Ionic Bonding

Net potential energy for a pair of oppositely charged ions

✓ E_{net} is minimum if ionic pair locates at an specific interionic distance a_0 . This is the equilibrium distance where attractive and repulsive energies are balanced.

$$F = \left(\frac{\partial E}{\partial a} \right) \Rightarrow E_{\text{net}} \Rightarrow$$

$$E_{\text{net}} = \underbrace{\frac{Z_1 Z_2 e^2}{4\pi \epsilon_0 a_o^2}}_{\text{Attraction Energy}} + \underbrace{\frac{b}{a_o^n}}_{\text{Repulsion Energy}}$$

Ionic Bonding

Lattice energy (U)

- ✓ Energy to bring a mol of ions from an infinite distance to fixed position in a crystal.

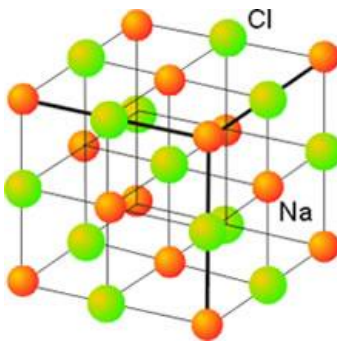
$$E_{r=a_0} = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 a_0} + \frac{b}{a_0^n}$$



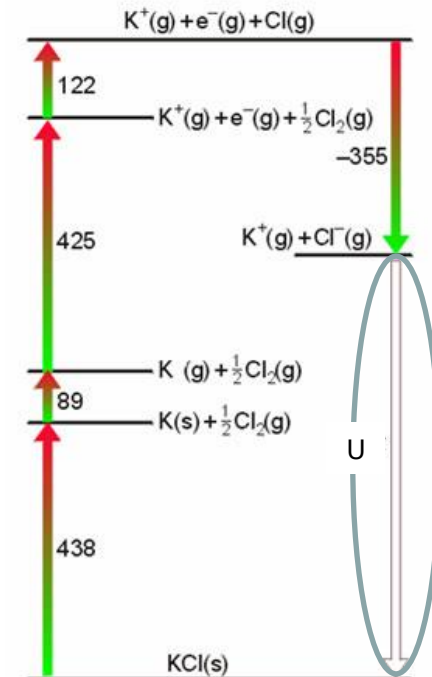
$$U = -\frac{Z_1 Z_2 e^2 N_A A}{4\pi\epsilon_0 a_0} \left(1 - \frac{1}{n}\right)$$

Lattice energy (Alternative view):

Energy released when a mole of ionic solid is formed from its ions in the gas state.



Step	ΔH (kJ/mol)
K sublimation	+89
$K_{(g)}$ ionization	+425
$Cl_{2(g)}$ dissociation	+244
$Cl_{(g)}$ ionization	-355
<i>KCl formation</i>	-438



Born-Haber cycle for KCl.

Ionic Bonding

Properties associated with Lattice energy (U):

1. Melting point

Ionic solids	Lattice energy		Melting Point °C
	kJ/mol	kcal/mol	
LiCl	829	198	613
NaCl	766	183	801
KCl	686	164	776
RbCl	670	160	715
CsCl	649	155	646
MgO	3932	940	2800
CaO	3583	846	2580
SrO	3311	791	2430
BaO	3127	747	1923

Valence +1

When the ion size $\uparrow \Rightarrow$ lattice energy $\downarrow$ and $T_m \downarrow$

Valence +2

Valence number $\uparrow \Rightarrow$ lattice energy $\uparrow \Rightarrow T_m \uparrow$

* All energy values are negative

Ionic Bonding

Properties associated with Lattice energy (U):

2. Hardness

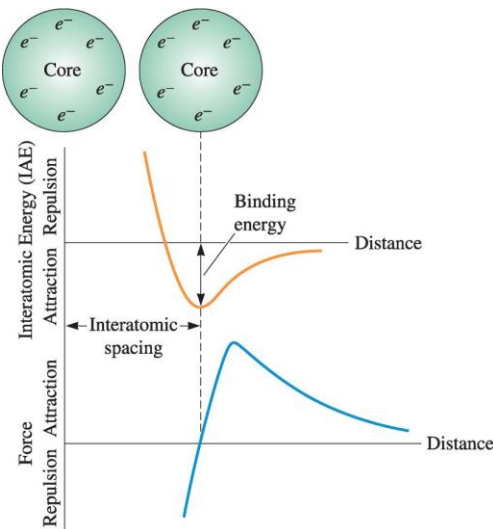
Hardness is a measure of the resistance to localized plastic deformation induced by either mechanical indentation or abrasion. Several scales, e.g. Mohs' scale (mineralogists)

	BeO	MgO	CaO	SrO	BaO
d (Å)	1,65	2,10	2,40	2,57	2,77
Mohs scale	9,0	6,5	4,5	3,5	3,3

$$r_{c+} \downarrow \Rightarrow \uparrow U \Rightarrow \text{hardness} \uparrow$$

Hydroxyapatite : 5

Keratin (fingernail): 2.5



The binding or potential energy, is the energy required to create or break the bond. Materials with high binding energy have high melting temperatures and hardness

Ionic Bonding

Properties associated with Lattice energy (U):

3. Thermal expansion is the tendency of matter to change in volume in response to a change in temperature.

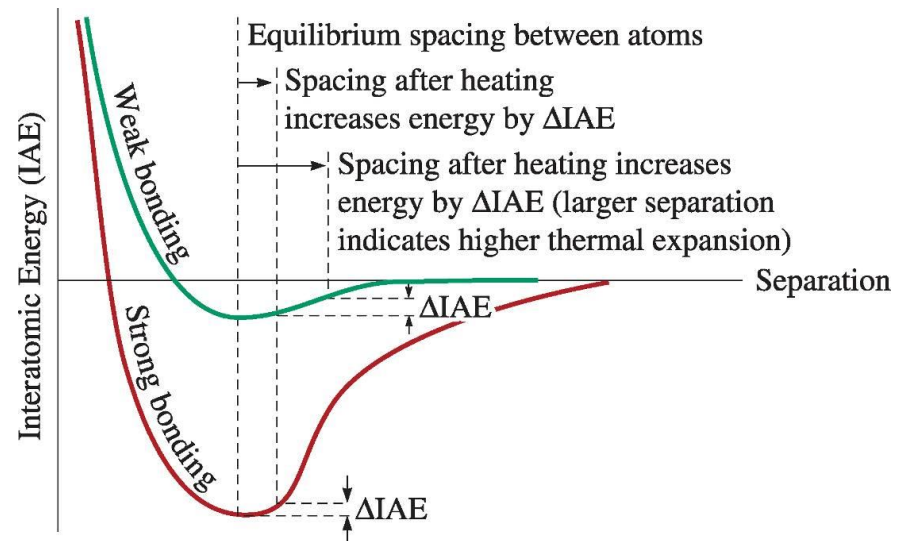
Coefficient of thermal expansion (change in linear dimension of a material per degree temperature): related to strength of atomic bonds.

Materials that display a steep curve with a **deep interatomic energy (IAE) trough**, caused by strong atomic bonding, have **low linear coefficients of thermal expansion**.

	...	NaF	NaCl	NaBr	NaI
$\alpha_v (10^6) (^{\circ}\text{C}^{-1})$	...	39	40	43	48

Hydroxyapatite : $12 \cdot 10^{-6} ^{\circ}\text{C}^{-1}$

$r_a \uparrow \Rightarrow \downarrow U \Rightarrow \text{thermal expansion } (\alpha) \uparrow$

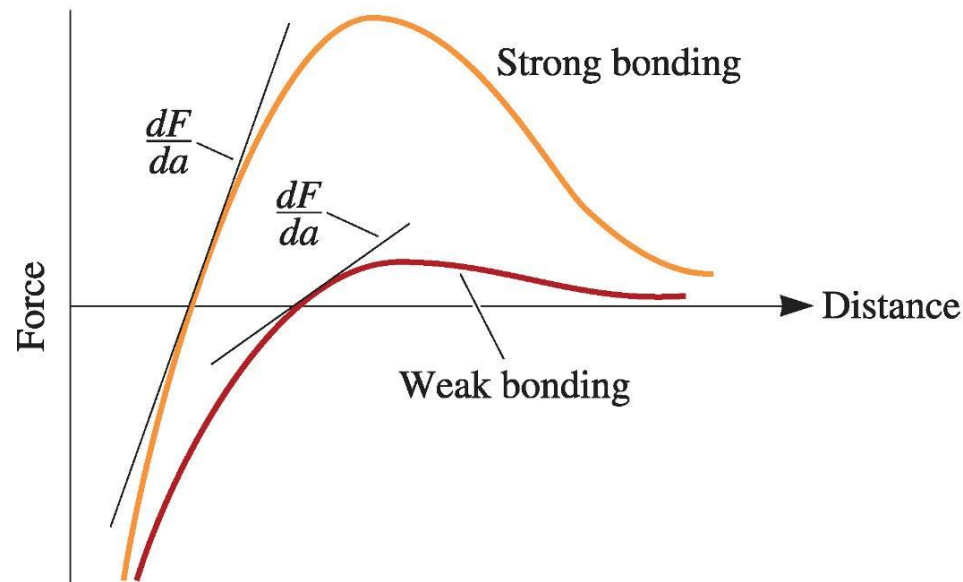


Ionic Bonding

Properties associated with Lattice energy (U):

2. Youngs Modulus

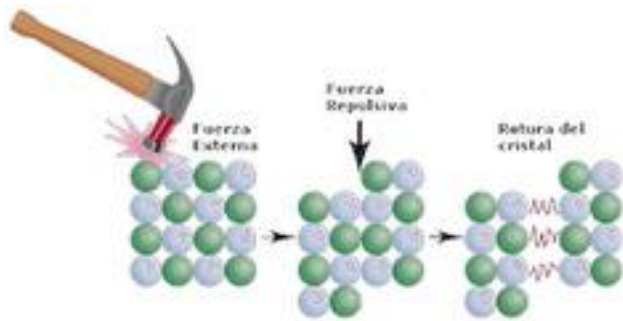
A high binding energy (steep slope), greater force required to stretch the bonds; thus **higher elastic (Young's) modulus**.



Ionic Bonding

- General properties of ionic compounds

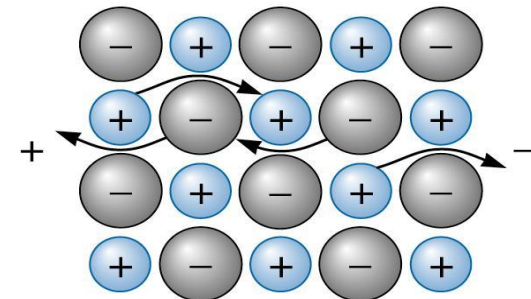
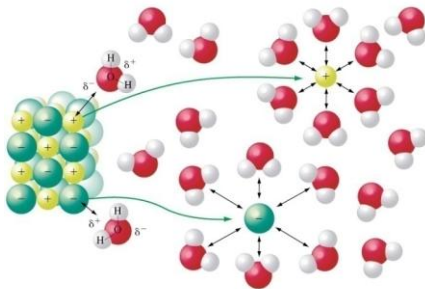
1. Strong electrostatic attraction → High melting and evaporation points
2. Hard and brittle solids at room temperature



3. They do not conduct electricity (except in molten state or when dissolved in water)

- When voltage is applied to an ionic material, entire ions must move to cause a current to flow. Ion movement is slow and the electrical conductivity is poor

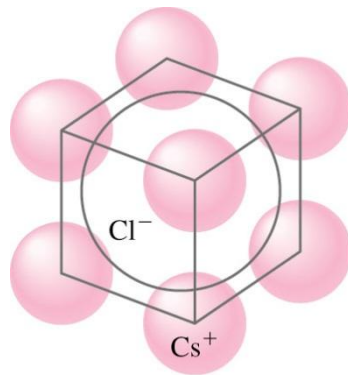
4. Water soluble



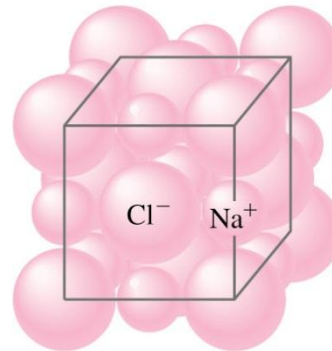
Ionic Bonding

Ion Arrangements in Ionic Solids

- Ionic bonds are ***Non Directional***
- Geometric arrangements are present in solids to maintain electric neutrality.



CsCl



NaCl

Ionic packing
In NaCl
and CsCl

- As the ratio of cation to anion radius decreases, fewer anion surround central cation.
- Ionic compounds are crystalline solids

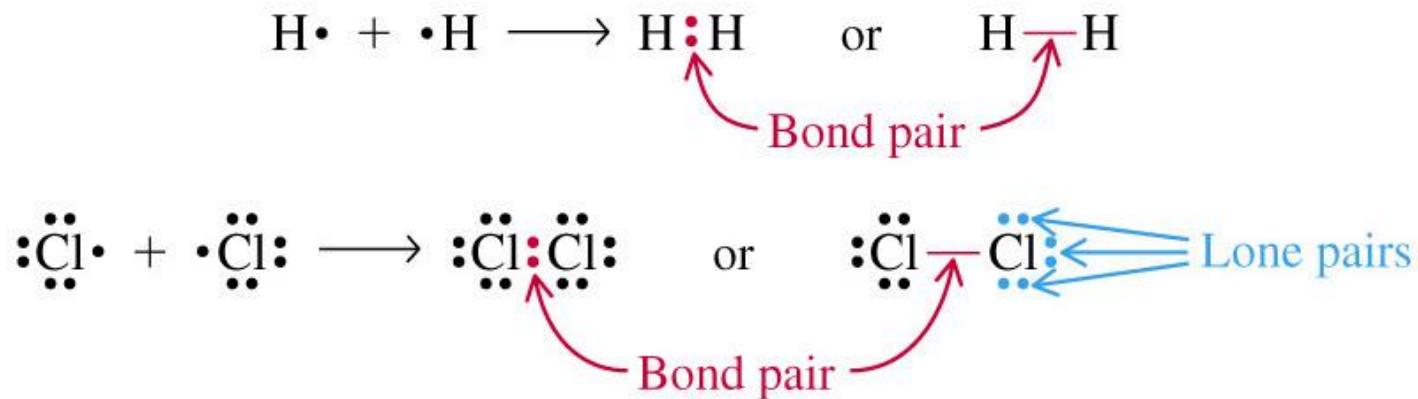
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Covalent Bonding

- Generally it forms between the non metallic elements of the periodic table.
- It forms by electron sharing.
- Highly directional. No free electrons
- Takes place between elements with small differences in electronegativity and close by in periodic table.
- In Covalent bonding, outer s and p electrons are shared between two atoms to obtain noble gas configuration.

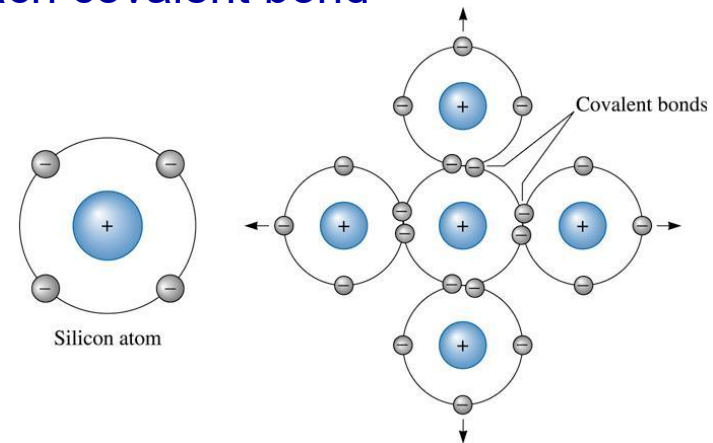
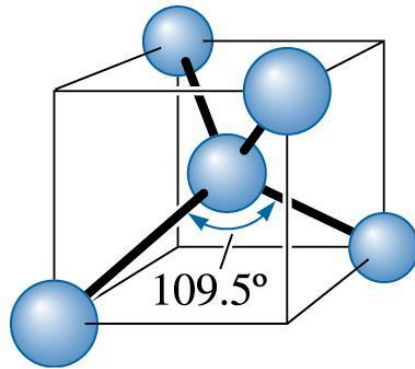
Examples:



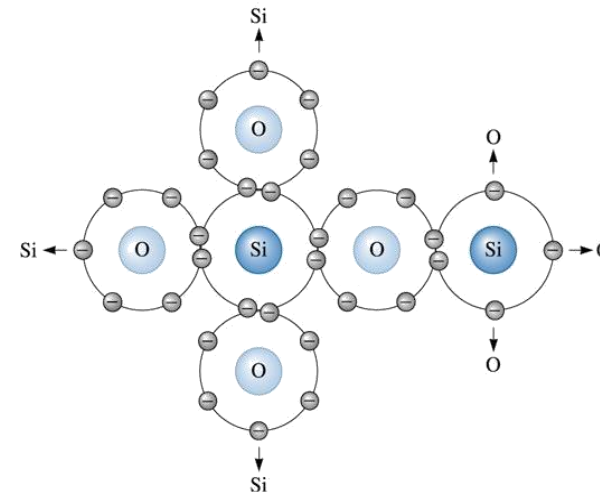
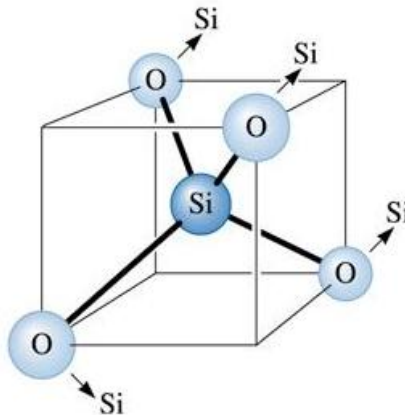
Covalent Bonding

Examples:

In **silicon**, with a valence of four, four covalent bonds must be formed. Covalent bonds are **directional**. In silicon, a tetrahedral structure is formed, with angles of 109.5° required between each covalent bond



The tetrahedral structure of **silica (SiO_2)**, which contains covalent bonds between silicon and oxygen atoms



Covalent Bonding

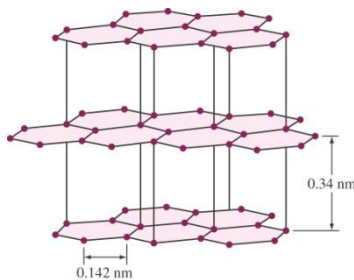
- Properties of the compounds with covalent bonds

- ✓ Covalent Solids

Formed by a system of continuous covalent bonds

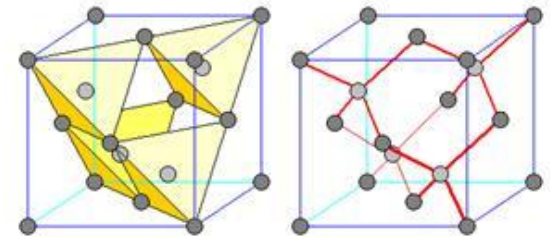
Non conductive LATTICES both in the solid and in the molten state

Diamond, boron nitride, quartz (SiO_2), silicon carbide (SiC)



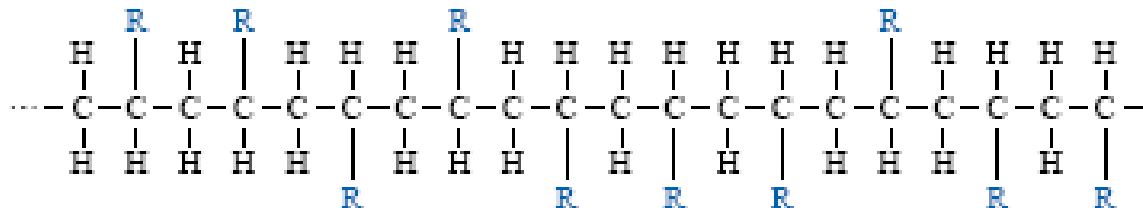
graphite

- ✓ Hard and incompressible
- ✓ T_m high, non volatile
- ✓ insoluble



Diamond

- ✓ Substances with covalent bonds

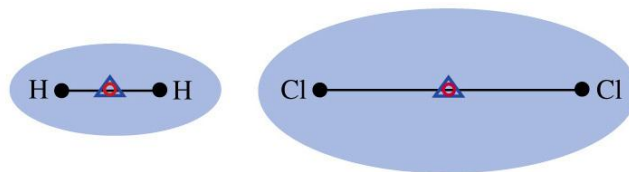


Covalent Bonding

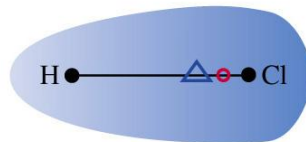
POLAR COVALENT BOND

A **polar covalent bond** exists when atoms with different electronegativities share electrons in a covalent bond.

Nonpolar bonds: sharing of the bonding electrons is equal between the atoms and the bonds.



(a) Nonpolar covalent bonds



(b) Polar covalent bond

- = Atomic nucleus
- △ = Center of positive charge
- = Center of negative charge

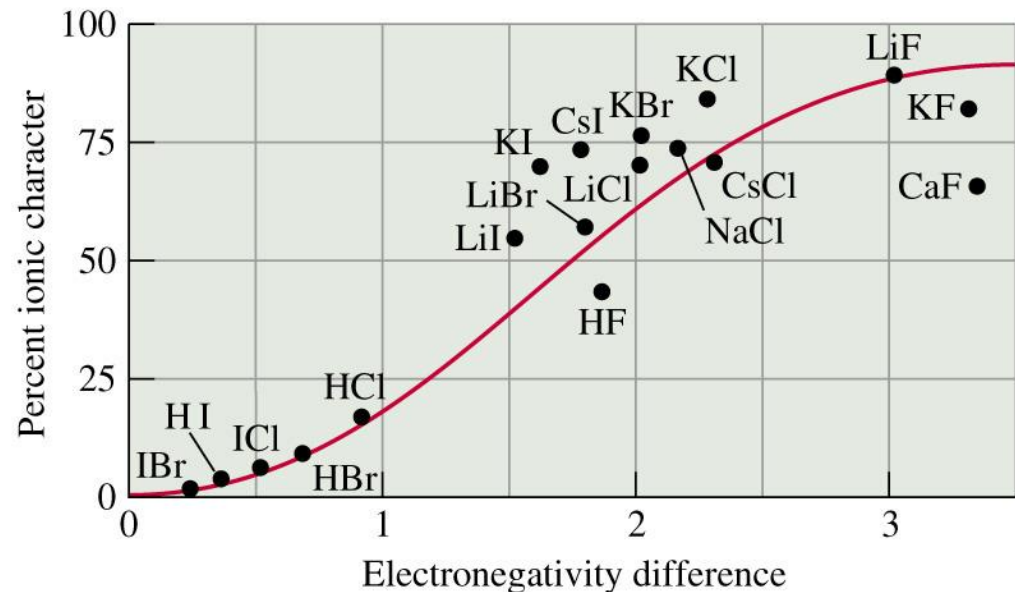
Dipole moment

$$\mu = Q \times d$$

d = Distance between atoms;

Q = charge

Units: Debye, D; $1 \text{ D} = 3.33 \times 10^{-30} \text{ C}\cdot\text{m}$



$$\% \text{ ionic character} = \left(1 - e^{-0.25(X_A - X_B)^2}\right) \times 100\%$$

X_i : electronegativity of element i

Covalent Bonding

Bond Length, Bond order and Bond Energy

Bond energies and lengths of selected covalent bonds

Bond	Bond energy*		Bond length (nm)
	kcal/mol	kJ/mol	
C—C	88	370	0.154
C=C	162	680	0.13
C≡C	213	890	0.12
C—H	104	435	0.11
C—N	73	305	0.15
C—O	86	360	0.14
C=O	128	535	0.12
C—F	108	450	0.14
C—Cl	81	340	0.18
O—H	119	500	0.10
O—O	52	220	0.15
O—Si	90	375	0.16
N—O	60	250	0.12
N—H	103	430	0.10
F—F	38	160	0.14
H—H	104	435	0.074

For a given pair of atoms, with higher bond order, the bond length will decrease; as bond length decreases, bond energy will increase (H₂, F₂, N₂)

*Approximate values since environment changes energy. All values are negative for bond formation (energy is released).

Source: L. H. Van Vlack, "Elements of Materials Science," 4th ed., Addison-Wesley, 1980.

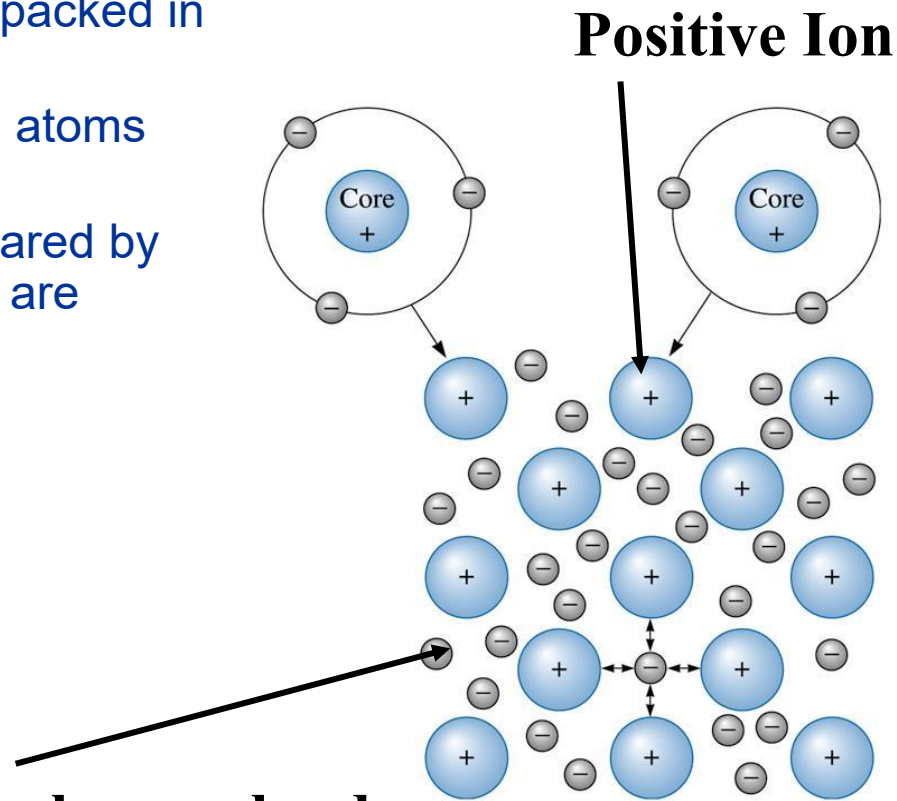
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Metallic Bonding

The metallic bond forms when atoms give up their valence electrons, which then form an **electron sea**. The positively charged atom cores are bonded by mutual attraction to the negatively charged electrons. Metallic bonds result from the electrostatic attraction between metal cations and delocalized electrons.

- Atoms in metals are closely packed in crystal structure.
- Electrons spread out among atoms forming **electron clouds**.
- Since outer electrons are shared by many atoms, metallic bonds are **Non-directional**



Metallic Bonding

- Overall energy of individual atoms are lowered by metallic bonds
- Minimum energy between atoms exist at equilibrium distance a_0
- Fewer the number of **valence electrons** involved, lower is the bonding energy.
 - Example:- $K \rightarrow +1$ valance $\rightarrow$ Bonding energy 89.6KJ/mol,
 $\rightarrow$ Melting temperature 63.5°C
- Higher the number of **valence electrons** involved, higher is the bonding energy.
 - Example:- $Ca \rightarrow +2$ valance $\rightarrow$ Bonding energy 177KJ/mol,
 $\rightarrow$ Melting temperature 851°C

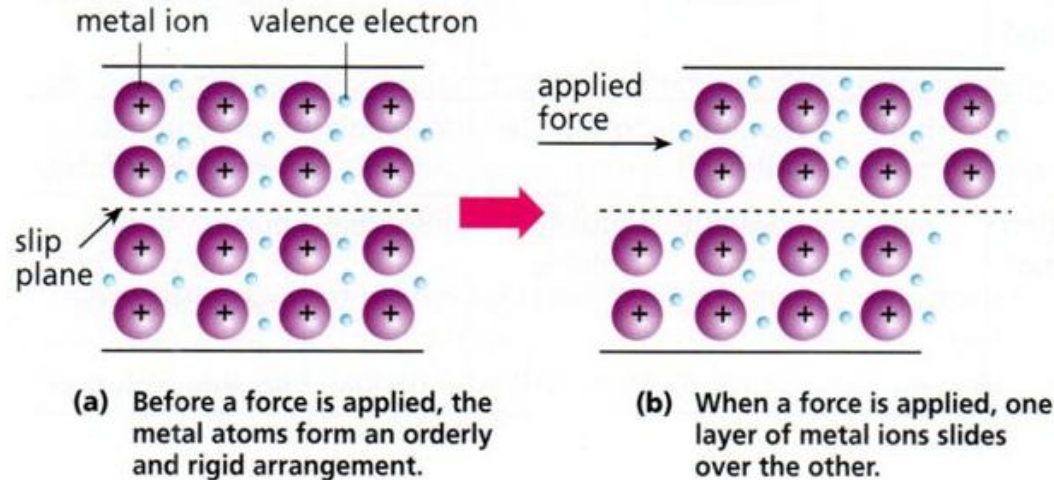
Bonding energies, melting points, and electron configurations of the fourth-period metals of the periodic table

Element	Electron configuration	Bonding energy		Melting point (°C)
		kJ/mol	kcal/mol	
K	$4s^1$	89.6	21.4	63.5
Ca	$4s^2$	177	42.2	851
Sc	$3d^1 4s^2$	342	82	1397
Ti	$3d^2 4s^2$	473	113	1660
V	$3d^3 4s^2$	515	123	1730
Cr	$3d^5 4s^1$	398	95	1903
Mn	$3d^5 4s^2$	279	66.7	1244
Fe	$3d^6 4s^2$	418	99.8	1535
Co	$3d^7 4s^2$	383	91.4	1490
Ni	$3d^8 4s^2$	423	101	1455
Cu	$3d^{10} 4s^1$	339	81.1	1083
Zn	$4s^2$	131	31.2	419
Ga	$4s^2 4p^1$	272	65	29.8
Ge	$4s^2 4p^2$	377	90	960

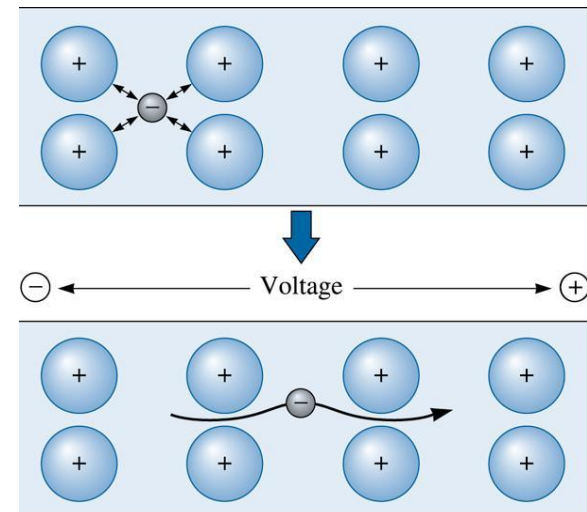
Metallic Bonding

Metallic Bonds and Material Properties

- Pure metals are significantly more malleable than ionic or covalent networked materials. Strength of a pure metal can be significantly increased through alloying.



- Pure metals are excellent conductors of heat and electricity. When voltage is applied to a metal, the electrons in the electron sea can easily move and carry a current



Metallic Bonding

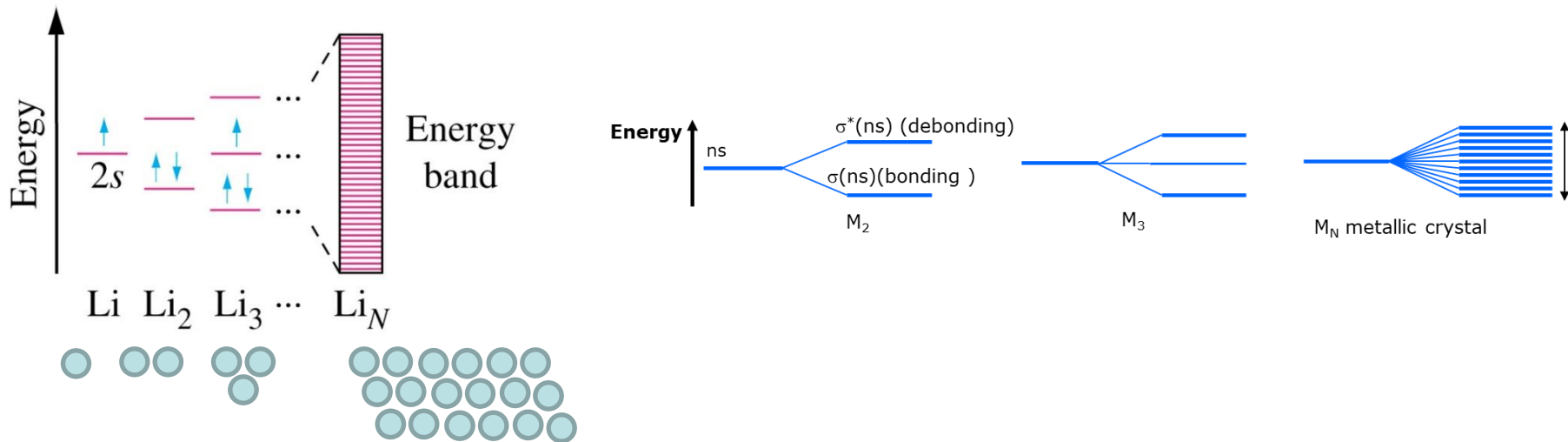
Band theory

Theoretical model describing the states of electrons, in solid materials, that can have values of energy only within certain specific ranges.

All Molecular Orbitals (MO) of free atoms transform in a group of non localized set of orbitals in a metallic crystal called BAND. Difference in energy between adjacent MOs is so small that energy is almost a continuum.

Metals can be considered as giant molecules formed by a giant number of MO's

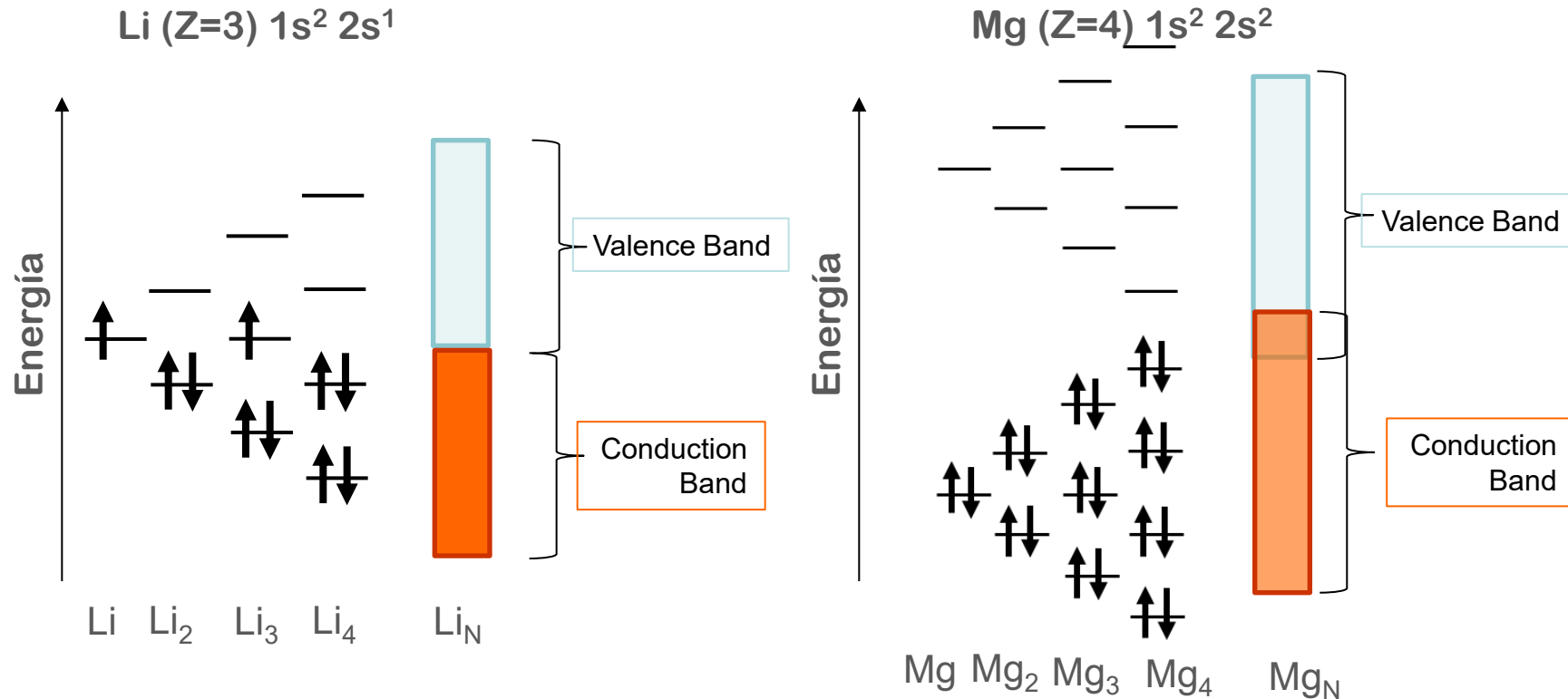
Li: $[1s^2 2s^1]$



Metallic Bonding

Band theory

Band structure depends on interatomic distance $\rightarrow$ outermost atomic orbitals overlap forming an energy band.



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Intermolecular Forces

Primary bonds: (strong)

-Ionic bond:
(150-370 kcal/mol)

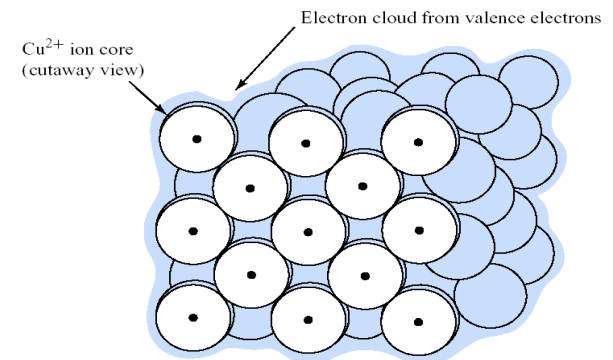
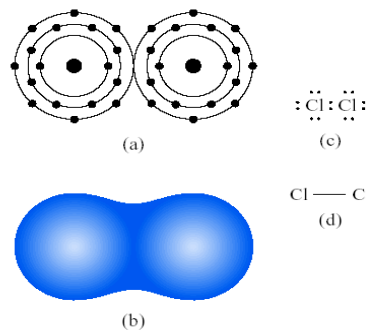
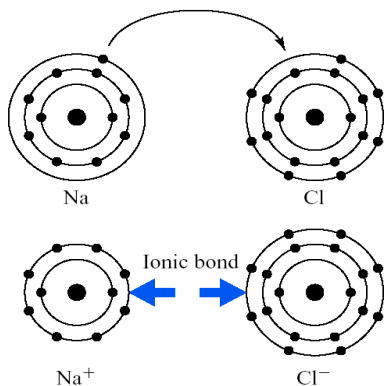
Transfer of $e^- \Rightarrow$ cations and anions $\Rightarrow F_{\text{coulomb}}$ (non directional)

- Covalent bond:
(125-300 kcal/mol)

Sharing e^- (directional)

- Metallic bond:
(25-200 kcal/mol)

e^- shared externally and very little tightened by the nucleus (non directional)



SHACKELFORD, J.F. "Introduction to materials science for engineers",

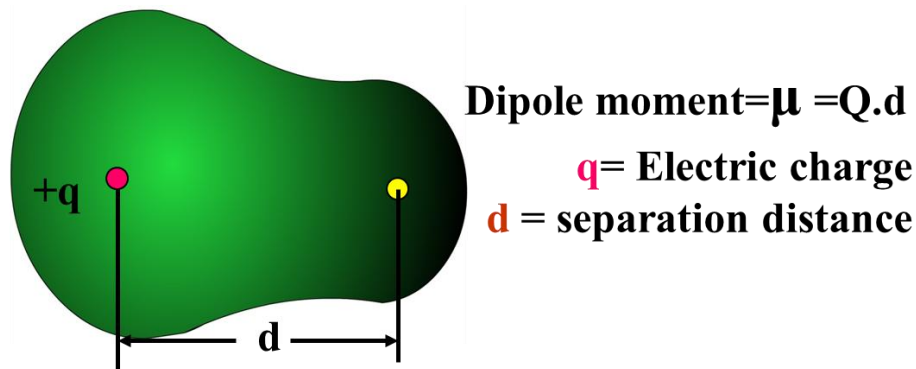
Secondary Bonding- Intermolecular Forces

Secondary bonds - weak (Van der Waals)

Secondary bonds are due to attractions of **electric dipoles** in atoms or molecules. Dipoles are created due to *asymmetrical distribution* of electron charges and a different positive and negative charge centers exist.

There are two types of dipoles:

- **Permanent dipoles:** do not fluctuate with time
- **Fluctuating dipoles:** Electron cloud charge changes with time.

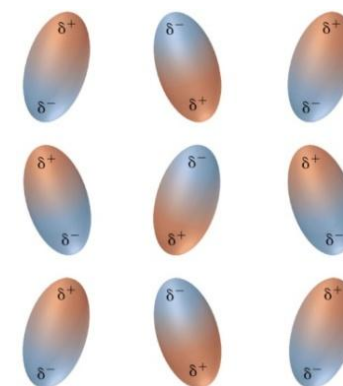


Secondary Bonding- Intermolecular Forces

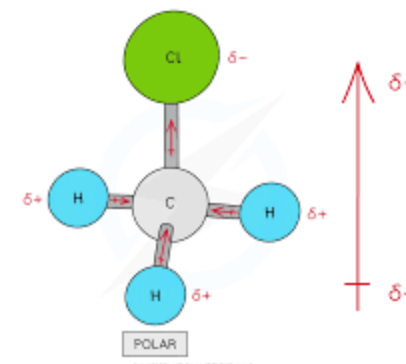
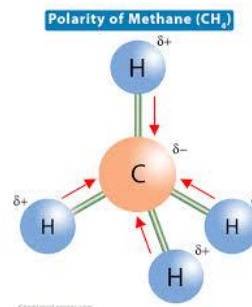
Secondary bonds - weak (Van der Waals)

1. Permanent dipole 1-10 kcal/mol It forms in molecules:

- that present constant dipole moment μ
- made of atoms with different electronegativity that are united with covalent bonding.



Example :

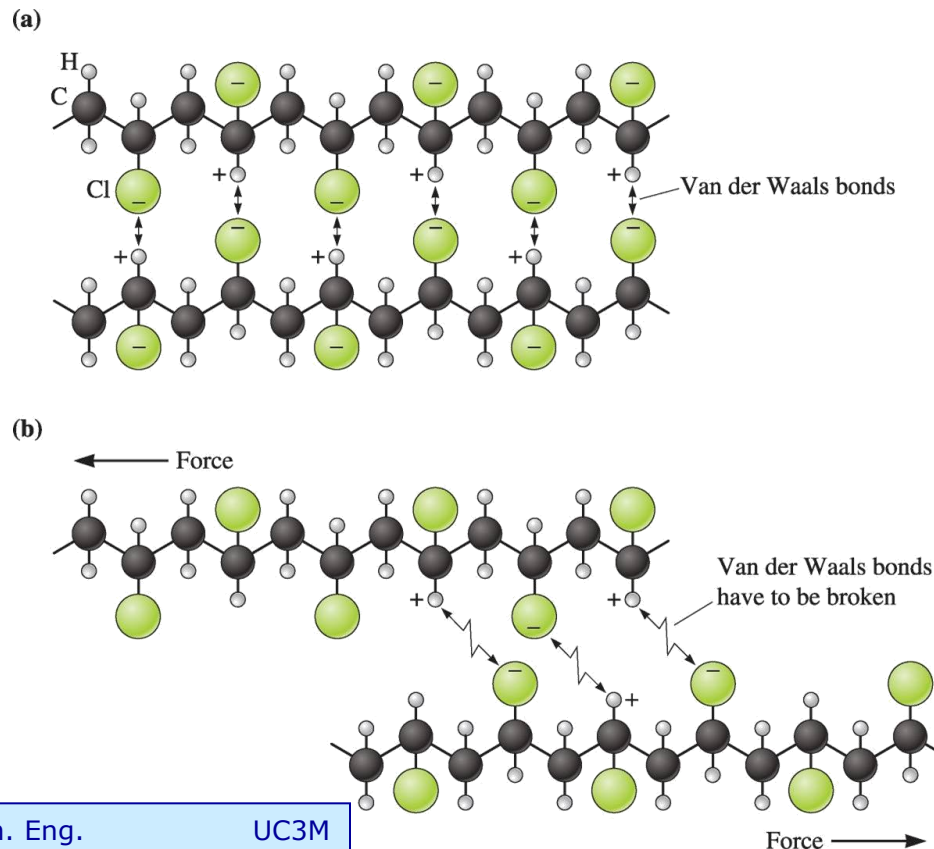


Secondary Bonding- Intermolecular Forces

Secondary bonds - weak (Van der Waals)

Example :

- (a) In polyvinyl chloride (PVC), the chlorine atoms attached to the polymer chain have a negative charge and the hydrogen atoms are positively charged. The chains are weakly bonded by van der Waals bonds. This additional bonding makes PVC stiffer,
- (b) When a force is applied to the polymer, the van der Waals bonds are broken and the chains slide past one another

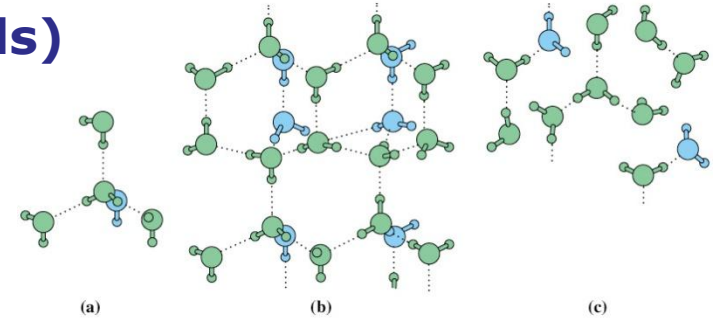


Secondary Bonding- Intermolecular Forces

Secondary bonds - weak (Van der Waals)

1. Permanent dipole

Hydrogen bond 7 kcal/mol in H_2O :

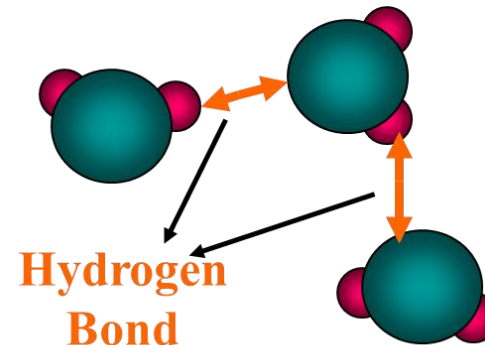
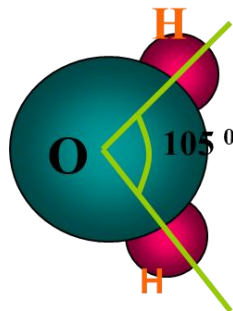


In molecules with H and **electronegative atoms** (O-H, N-H, F-H) there is an asymmetric distribution of charge $\Rightarrow$ Polar covalent bond $\Rightarrow$ Permanent dipole

- Hydrogen bonds are *Dipole-Dipole interaction* between polar bonds containing hydrogen atom

Example :-

- ❖ In water, dipole is created due to asymmetrical arrangement of hydrogen atoms.
- ❖ Attraction between positive oxygen pole and negative hydrogen pole.

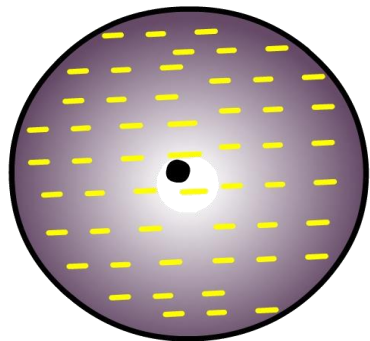
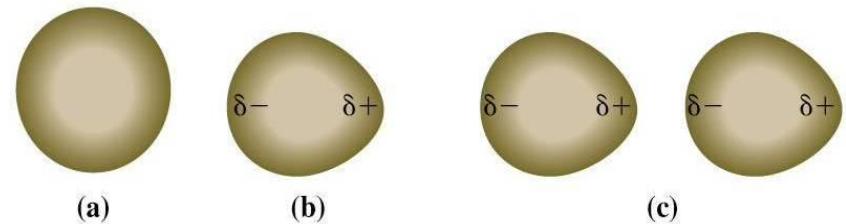


Secondary Bonding- Intermolecular Forces

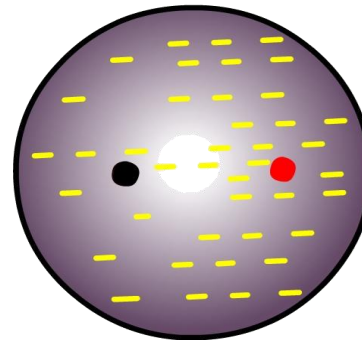
Secondary bonds - weak (Van der Waals)

2. Fluctuating dipole: $< 2 \text{ kcal/mol}$. It forms in non polar molecules in crystalline lattice

• Instantaneous fluctuations of electron charge distribution $\Rightarrow$ *fluctuating dipoles*



**Symmetrical
distribution
of electron charge**



**Asymmetrical
Distribution
(Changes with time)**

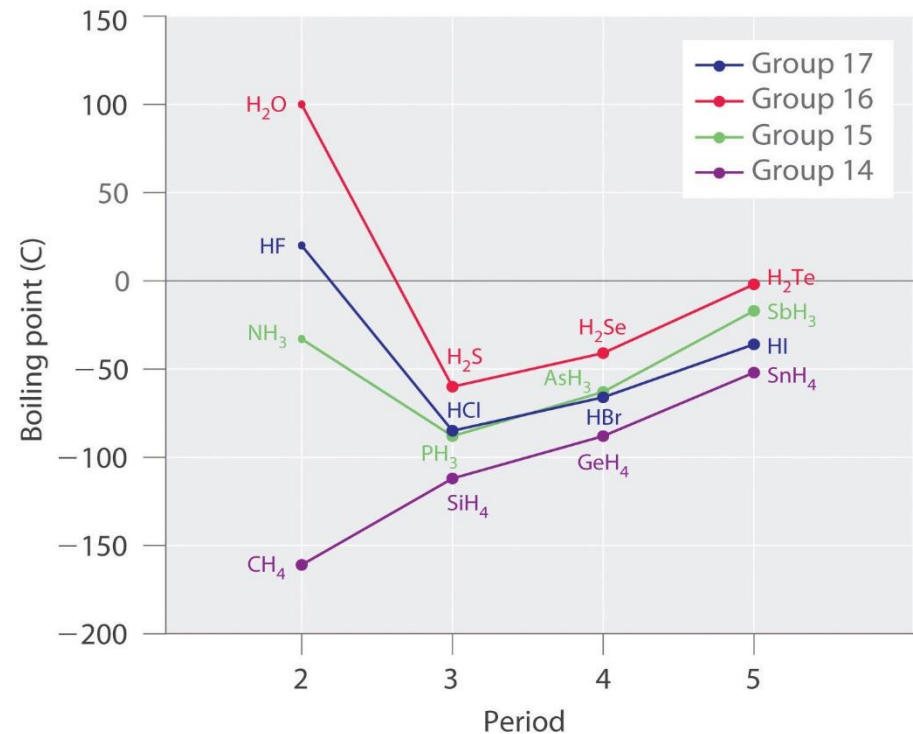
Secondary Bonding- Intermolecular Forces

Secondary bonds (Van der Waals):

Example

Elements in higher periods generally have higher melting points because atoms in higher periods share more electrons, leading to stronger forces that hold the atoms or molecules together in a solid state.

However, the presence of strong hydrogen bonding results in a significantly higher boiling point in H_2O , HF , NH_3 .



Variation of melting points of molecular hydrides

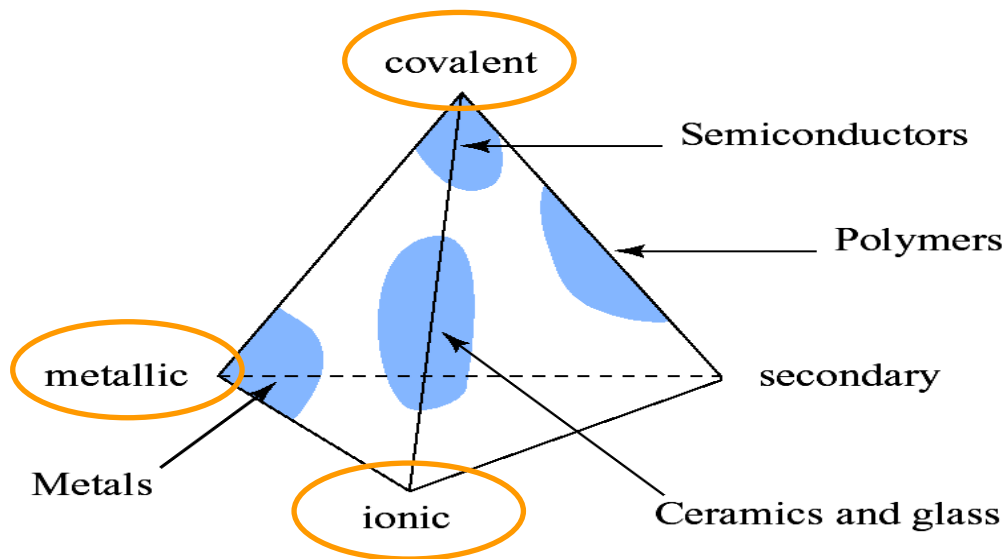
TOPIC 1. BONDING IN SOLIDS

- Introduction
- Bonding in solids
 - Ionic bonding
 - Covalent bonding
 - Metallic bonding
 - Secondary bonding
- **Bonding and properties in materials**

Bonding and properties in materials

Nature of bonding ↔ properties ↔ type of materials

Type of material	Character of bonding	Examples
Metal	Metallic	Fe, steels
Ceramic and glasses	Ionic/ covalent	Silica (SiO_2)
Polymers	Covalent and secondary	Polyethylene $-(\text{CH}_2)-$



Material	Bonding type	Melting point ($^{\circ}\text{C}$)
NaCl	Ionic	801
C (diamond)	Covalent	~ 3550
$-(\text{C}_2\text{H}_4)-_n$	Covalent and secondary	$\sim 120^a$
Cu	Metallic	1084.87
Ar	Secondary (induced dipole)	-189
H_2O	Secondary (permanent dipole)	0

^aBecause of the irregularity of the polymeric structure of polyethylene, it does not have a precise melting point. Instead, it softens with increasing temperature above 120°C . In this case, the 120°C value is a “service temperature” rather than a true melting point.

Bonding and properties in materials

Table 2.3 Bonding Energies and Melting Temperatures for Various Substances

Bonding Type	Substance	Bonding Energy		Melting Temperature (°C)
		<i>kJ/mol</i>	<i>eV/Atom, Ion, Molecule</i>	
Ionic	NaCl	640	3.3	801
	MgO	1000	5.2	2800
Covalent	Si	450	4.7	1410
	C (diamond)	713	7.4	>3550
Metallic	Hg	68	0.7	−39
	Al	324	3.4	660
	Fe	406	4.2	1538
	W	849	8.8	3410
van der Waals	Ar	7.7	0.08	−189
	Cl ₂	31	0.32	−101
Hydrogen	NH ₃	35	0.36	−78
	H ₂ O	51	0.52	0

Bond	Type of substance	Melting and boiling points	Mechanical properties	Solubility
COVALENT	Molecular	Low	Soft in the solid state	Depends on the polarity of the molecules
COVALENT	Atomic, covalent or lattice	Very high	Very hard brittle	Insoluble in all solvents
IONIC	Ionic	High	Hard and brittle	Soluble in polar solvents
METALLIC	Metallic	High	Ductile and workable	Insoluble in all solvents

Questions

- Compare and contrast metallic and covalent primary bonds in terms of
 - a) the nature of the bond,
 - b) the valence of the atoms involved, and
 - c) the ductility of the materials bonded in these ways.
- Titanium is stiffer than aluminum, has a lower thermal expansion coefficient than aluminum, and has a higher melting temperature than aluminum. On the same graph, carefully and schematically draw the potential well curves for both metals and showing how the physical properties are manifested in these curves.
- Would you expect MgO or magnesium to have the higher modulus of elasticity? Explain.
- Aluminum and silicon are side-by-side in the periodic table. Which would you expect to have the higher modulus of elasticity (E)? *Explain.*